

ASSIGNMENT 01

Title : Installation and Simulation of Cloud Computing in CloudSim.

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Code (CloudSimExample1.java) :-

```
import java.text.DecimalFormat;
import java.util.*;

import org.cloudbus.cloudsim.*;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.*;

public class CloudSimExample1Short {

    public static void main(String[] args) {
        Log.println("Starting CloudSimExample1...");

        try {
            CloudSim.init(1, Calendar.getInstance(), false);

            Datacenter dc = createDatacenter("Datacenter_0");
            DatacenterBroker broker = new DatacenterBroker("Broker");
            int brokerId = broker.getId();

            Vm vm = new Vm(0, brokerId, 1000, 1, 512, 1000, 10000, "Xen",
                new CloudletSchedulerTimeShared());
            broker.submitVmList(Arrays.asList(vm));

            UtilizationModel um = new UtilizationModelFull();
            Cloudlet c1 = new Cloudlet(0, 400000, 1, 300, 300, um, um, um);
            Cloudlet c2 = new Cloudlet(1, 400000, 1, 300, 300, um, um, um);
            c1.setUserId(brokerId);
            c2.setUserId(brokerId);

            broker.submitCloudletList(Arrays.asList(c1, c2));

            CloudSim.startSimulation();
            List<Cloudlet> result = broker.getCloudletReceivedList();
            CloudSim.stopSimulation();

            printResults(result);
            Log.println("CloudSimExample1 finished!");

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

```

static Datacenter createDatacenter(String name) throws Exception {
    List<Pe> peList = Arrays.asList(new Pe(0, new PeProvisionerSimple(1000)));

    Host host = new Host(0, new RamProvisionerSimple(2048),
        new BwProvisionerSimple(10000), 1000000, peList,
        new VmSchedulerTimeShared(peList));

    DatacenterCharacteristics ch = new DatacenterCharacteristics(
        "x86", "Linux", "Xen",
        Arrays.asList(host), 10.0, 3.0, 0.05, 0.001, 0.0);

    return new Datacenter(name, ch,
        new VmAllocationPolicySimple(Arrays.asList(host)),
        new LinkedList<Storage>(), 0);
}

static void printResults(List<Cloudlet> list) {
    DecimalFormat df = new DecimalFormat("###.##");

    Log.println("\n===== OUTPUT =====");
    Log.println("Cloudlet ID   STATUS   Data center ID   VM ID   Time   Start Time
Finish Time");

    for (Cloudlet c : list) {
        if (c.getCloudletStatus() == Cloudlet.SUCCESS) {
            Log.println("   " + c.getCloudletId()
                + "      SUCCESS      "
                + c.getResourceId()
                + "      " + c.getVmId()
                + "      " + df.format(c.getActualCPUTime())
                + "      " + df.format(c.getExecStartTime())
                + "      " + df.format(c.getFinishTime()));
        }
    }
}
}

```

OUTPUT :-

Starting CloudSimExample1...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 1 resource(s)
0.0: Broker: Trying to Create VM #0 in Datacenter_0
0.1: Broker: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker: Sending cloudlet 0 to VM #0
0.1: Broker: Sending cloudlet 1 to VM #0
400.1: Broker: Cloudlet 0 received
800.1: Broker: Cloudlet 1 received
800.1: Broker: All Cloudlets executed. Finishing...
800.1: Broker: Destroying VM #0
Broker is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_0 is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.

===== OUTPUT =====

Cloudlet ID	STATUS	Data center ID	VM ID	Time	Start Time	Finish Time
0	SUCCESS	2	0	400	0.1	400.1
1	SUCCESS	2	0	400	400.1	800.1

CloudSimExample1 finished!
BUILD SUCCESSFUL (total time: 0 seconds)